Midterm Exam Solutions:

1. **(Conditional Probability; 25 pts)**
Suppose you have three email accounts, with the following properties:

Account	Messages Received (% of **inbox**)	Spam (% of each **account**)
1	70%	1%
2	20%	2%
3	10%	5%

We randomly select one email from our inbox, such that all emails are equally likely to be selected.

(a) What is the probability that this email is from Account 1?

(b) What is the probability that this email is spam?

(c) What is the conditional probability that this email is from Account 1, *given that it is spam*?

(d) What is the conditional probability that this email is from Account 1, *given that it is **not** spam*?

(e) Define two events: $A = \{\text{selected email is from Account 1}\}$, and $B = \{\text{selected email is spam}\}$. Are the two events A and B independent? Justify your answer.

a) $\mathbb{P}(\text{Account } 1) = \dfrac{\#\text{ emails in Acc. } 1}{\#\text{ emails in inbox}}$ by equally likely outcomes

$\boxed{\mathbb{P}(\text{Account } 1) = \underline{70\%}}$ by given information

b) Use the Law of Total Probability:

$$\mathbb{P}(\text{Spam}) = \mathbb{P}(\text{Spam} \mid \text{Acc. } 1) \cdot \mathbb{P}(\text{Acc. } 1) + \mathbb{P}(\text{Spam} \mid \text{Acc. } 2) \cdot \mathbb{P}(\text{Acc. } 2)$$

$$+ \mathbb{P}(\text{Spam} \mid \text{Acc. } 3) \cdot \mathbb{P}(\text{Acc. } 3)$$

$$= 1\% \cdot 70\% + 2\% \cdot 20\% + 5\% \cdot 10\%$$

since, e.g. $\mathbb{P}(\text{Spam} \mid \text{Acc. } 1) = 1\%$ (and other given quantities)

$\boxed{\mathbb{P}(\text{Spam}) = \underline{1.6\%}}$

c) Use Bayes' Rule:

$$\mathbb{P}(\text{Acc.}1 \mid \text{Spam}) = \frac{\mathbb{P}(\text{Spam} \mid \text{Acc.}1) \cdot \mathbb{P}(\text{Acc.}1)}{\mathbb{P}(\text{Spam})}$$

$$= \frac{1\% \cdot 70\%}{1.6\%}$$ plugging in from b) and givens

$$\boxed{\mathbb{P}(\text{Acc.}1 \mid \text{Spam}) = 43.75\%}$$

d) Use Bayes' Rule:

$$\mathbb{P}(\text{Acc.}1 \mid \text{Not Spam}) = \frac{\mathbb{P}(\text{Not Spam} \mid \text{Acc.}1) \cdot \mathbb{P}(\text{Acc.}1)}{\mathbb{P}(\text{Not Spam})}$$

$$= \frac{(1 - \mathbb{P}(\text{Spam} \mid \text{Acc.}1)) \cdot \mathbb{P}(\text{Acc.}1)}{1 - \mathbb{P}(\text{Spam})}$$ by prob.'s of complements

$$= \frac{(1 - 1\%) \cdot 70\%}{1 - 1.6\%}$$ plugging in

$$\boxed{\mathbb{P}(\text{Acc.}1 \mid \text{Not Spam}) = 70.4\%}$$

e) $\mathbb{P}(A) = 70\%$
$\mathbb{P}(B) = 1.6\%$ from a) and b)

$$\mathbb{P}(A \cap B) = \mathbb{P}(B \mid A) \cdot \mathbb{P}(A)$$

$$= \mathbb{P}(\text{Spam} \mid \text{Acc.}1) \cdot \mathbb{P}(\text{Acc.}1)$$

$$= 1\% \cdot 70\% = 0.7\%$$

Since $\mathbb{P}(A) \cdot \mathbb{P}(B) = 1.1\% \neq 0.7\% = \mathbb{P}(A \cap B)$,
then events A and B are <u>not</u> independent!

2. **(Discrete Random Variables; 25 pts)**
You and a friend are playing a simple betting game. Given a fair coin (which is equally likely to land heads- or tails-up), you flip it however many times needed until the first heads appears, and record the total number of flips as the random variable X.

(a) What is the distribution of X? Be sure to specify: i) the family of distribution, ii) the relevant parameters, and iii) the probability mass function of X, $p_X(\cdot)$.

(b) Compute $\mathbb{P}(X > k)$ for $k = 1, 2, \ldots$.

(c) Define a new random variable Y, which represents your payoff in this game.

$$Y = \begin{cases} 10\$ & \text{if } X \leq 3 \\ -1\$ & \text{otherwise.} \end{cases}$$

Compute $p_Y(\cdot)$, the probability mass function of Y.

(d) Compute the expected payoff of this game, $\mathbb{E}[Y]$.

(e) Let us re-define the payoff random variable Y to be:

$$Y = \begin{cases} 10\$ & \text{if } X \leq 3 \\ -c\$ & \text{otherwise.} \end{cases}$$

for some constant c.
For what value of c is this game 'fair'? That is, what value of c yields $\mathbb{E}[Y] = 0$?

a) $X \sim \text{Geometric}(1/2)$, with pmf:

$$p_X(x) = \begin{cases} (1-\frac{1}{2})^{x-1} \cdot \frac{1}{2} & \text{for } x = 1, 2, 3, \ldots \\ 0 & \text{otherwise} \end{cases}$$

b) $\mathbb{P}(X > k) = \mathbb{P}(\text{Flips } 1, \ldots k \text{ are all failures, i.e. not heads})$

$= \mathbb{P}(\text{Flip } 1 \text{ is } F) \cdot \ldots \cdot \mathbb{P}(\text{Flip } k \text{ is } F)$ by mutual independence

$$\boxed{\mathbb{P}(X > k) = (1/2)^k, \text{ for } k = 1, 2, \ldots}$$

c) $\mathbb{P}(Y = -1) = \mathbb{P}(X > 3)$ by def'n of Y

$\mathbb{P}(Y = -1) = (1/2)^3$

$\mathbb{P}(Y = 10) = 1 - \mathbb{P}(Y = 0)$ since $Y=$ or $Y=$

$\mathbb{P}(Y = 10) = 1 - (1/2)^3$

$$\boxed{\text{So, } p_Y(y) = \begin{cases} 12.5\% & \text{if } y = -1 \\ 87.5\% & \text{if } y = 10 \\ 0 & \text{otherwise} \end{cases}}$$

d) $\mathbb{E}[Y] = (-1)\cdot(12.5\%) + (10)\cdot(87.5\%)$

$$\boxed{\mathbb{E}[Y] = 8.625}$$

e) Using parts c) and d), now:

$$\mathbb{E}[Y] = (-c)\cdot(12.5\%) + (10)\cdot(87.5\%)$$

Solve for c:

$$0 = -c \cdot 12.5\% + 10 \cdot 87.5\%$$

$$\boxed{\Rightarrow c = 70 \text{ implies that } \mathbb{E}[Y] = 0}$$

3. **(Continuous Random Variables; 25 pts)**
The Pareto distribution, whose probability density function is parameterized by $\alpha > 1$, is given by:

$$f_X(x;\alpha) = \frac{\alpha}{x^{\alpha+1}}, \quad \text{for } x \geq 1. \tag{1}$$

Let $X \sim \text{Pareto}(\alpha)$. (i.e., the density of X is given by Eq. (1) for some value of α)

(a) Verity that $f_X(x;\alpha)$ is a valid probability distribution function.

(b) Compute the cumulative distribution function of X, $F_X(x;\alpha)$.

(c) Compute $\mathbb{E}[X]$. Recall that $\alpha > 1$.
Hint: this should depend on α.

(d) For $\alpha = 4$, compute $\text{Var}(X)$.

(e) Compute $\text{Var}(X)$ for general choice of $\alpha > 1$.

a) Check:

- $f_X(x;\alpha) \geq 0$ ✓

- $\displaystyle\int_{-\infty}^{\infty} f_X(x;\alpha)\,dx = \int_1^{\infty} \frac{\alpha}{x^{\alpha+1}}\,dx = \left[\alpha\cdot\left(\frac{-1}{\alpha}\right)\frac{1}{x^{\alpha}}\right]_{x=1}^{x=\infty}$

$$= -0 - -\frac{1}{1^{\alpha}} \quad \left(\text{since } \alpha > 1,\ \frac{1}{x^{\alpha}} \to 0 \text{ as } x \to \infty\right)$$

$$\int_{-\infty}^{\infty} f_X(x;\alpha)\,dx = \underline{1} \ ✓$$

Hence, $f_X(x;\alpha)$ for $\alpha > 1$ is a valid pdf.

b) $F_X(x;\alpha) = \int_{-\infty}^{x} f_X(u;\alpha)\,du$

$$= \int_{1}^{x} \frac{\alpha}{u^{\alpha+1}}\,du \quad \text{for } x \geq 1,\ 0 \text{ otherwise}$$

$$= \left[\alpha \cdot \left(\frac{-1}{\alpha}\right)\frac{1}{u^{\alpha}}\right]_{u=1}^{u=x}$$

$$= -\frac{1}{x^{\alpha}} - {}^{-}1$$

$$\boxed{F_X(x;\alpha) = 1 - \frac{1}{x^{\alpha}} \quad \text{if } x \geq 1,\ 0 \text{ otherwise}}$$

c) $\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f_X(x;\alpha)\,dx$

$$= \int_{1}^{\infty} x \cdot \frac{\alpha}{x^{\alpha+1}}\,dx = \int_{1}^{\infty} \frac{\alpha}{x^{\alpha}}\,dx$$

$$= \left[\alpha \cdot \frac{-1}{\alpha-1} \cdot \frac{1}{x^{\alpha-1}}\right]_{x=1}^{\infty}$$

$$= -\frac{\alpha}{\alpha-1}\cdot 0 - {}^{-}\frac{\alpha}{\alpha-1}\cdot 1 \quad \left(\text{since } \alpha > 1,\ \frac{1}{x^{\alpha-1}} \to 0 \text{ as } x \to \infty\right)$$

$$\boxed{\mathbb{E}[X] = \frac{\alpha}{\alpha-1}}$$

d) Start with $\mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x;\overset{\alpha=4}{4})\,dx$

$$= \int_{1}^{\infty} x^2 \cdot \frac{4}{x^5}\,dx$$

$$= \int_{1}^{\infty} 4 \cdot \frac{1}{x^3}\,dx$$

$$= 4 \cdot \left[-\frac{1}{2} \cdot \frac{1}{x^2}\right]_{1}^{\infty}$$

$\mathbb{E}[X^2] = 2$

Now: $Var(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

$$= 2 - \left(\frac{4}{4-1}\right)^2$$

$$\boxed{Var(X) = 2/9}$$

e) More generally:

$$\mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x;\alpha)\,dx$$

$$= \int_1^{\infty} x^2 \cdot \frac{\alpha}{x^{\alpha+1}}\,dx = \int_1^{\infty} \frac{\alpha}{x^{\alpha-1}}\,dx$$

$$= \left[\frac{-\alpha}{\alpha-2} \cdot \frac{1}{x^{\alpha-2}}\right]_1^{\infty}$$

$$\mathbb{E}[X^2] = \begin{cases} \frac{\alpha}{\alpha-2} & \text{if } \alpha > 2 \quad \left(\text{since } \frac{1}{x^{\alpha-2}} \to 0 \text{ as } x \to \infty\right) \\ \infty & \text{if } \alpha \in (1,2] \quad \left(\text{since } -\frac{\alpha}{\alpha-2} \cdot \frac{1}{x^{\alpha-2}} \to \infty \text{ as } x \to \infty\right) \end{cases}$$

Hence:

$$\boxed{Var(X) = \begin{cases} \frac{\alpha}{\alpha-2} - \left(\frac{\alpha}{\alpha-1}\right)^2 = \frac{\alpha}{(\alpha-2)(\alpha-1)^2} & \text{if } \alpha > 2 \\ \infty & \text{otherwise} \end{cases}}$$

4. **(Jointly Distributed Random Variables; 25 pts)**
Let X and Y be continuous random variables, whose joint probabiltiy density funciton is given by

$$f(x,y) = \begin{cases} c\,(x^2+y^3) & \text{for } x \in [0,1],\ y \in [0,1] \\ 0 & \text{otherwise.} \end{cases}$$

for some constant c.

(a) For what value of c is $f(x,y)$ a valid joint probability density function?
(b) Compute the marginal probability density functions of X and Y, denoted $f_X(x)$ and Y, $f_Y(y)$.
(c) Are X and Y independent random variables? Justify your answer.
(d) Compute $\mathrm{Cov}(X,Y)$, the covariance between X and Y.
(e) Compute $\rho(X,Y)$, the correlation between X and Y.

a) We need $c \geq 0$, otherwise $f(x,y)$ is not non-negative.

We also need:

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$$

So:

$$1 = \int_0^1\int_0^1 c\cdot(x^2+y^3)\,dx\,dy$$

$$= c\cdot\int_0^1 \left[\tfrac{1}{3}x^3 + xy^3\right]_{x=0}^{x=1} dy$$

$$= c\cdot\int_0^1 \tfrac{1}{3} + y^3\,dy$$

$$= c\cdot\left[\tfrac{1}{3}y + \tfrac{1}{4}y^4\right]_{y=0}^{y=1}$$

$$1 = c\cdot\left(\tfrac{1}{3}+\tfrac{1}{4}\right)$$

$\Rightarrow \underline{c = \tfrac{12}{7}}$ for $f(x,y)$ to be valid

b) $f_X(x) = \int_0^1 \frac{12}{7}(x^2+y^3)\,dy$

$= \frac{12}{7}\left[yx^2 + \frac{1}{4}y^4\right]_{y=0}^{y=1}$

$\underline{f_X(x) = \frac{12}{7}\left(x^2+\frac{1}{4}\right)}$, for $x \in [0,1]$

$f_Y(y) = \int_0^1 \frac{12}{7}(x^2+y^3)\,dx$

$= \frac{12}{7}\left[\frac{1}{3}x^3 + xy^3\right]_{x=0}^{x=1}$

$\underline{f_Y(y) = \frac{12}{7}\left(\frac{1}{3}+y^3\right)}$, for $y \in [0,1]$

c) No. $f_X(x)\cdot f_Y(y) = \left(\frac{12}{7}\right)^2\left(x^2+\frac{1}{4}\right)\left(\frac{1}{3}+y\right) \neq \frac{12}{7}(x^2+y^3) = f(x,y)$

So X and Y are <u>not independent</u>.

d) First:

$\mathbb{E}[X] = \int_0^1 x\cdot f_X(x)\,dx = \int_0^1 \frac{12}{7}\left(x^3+\frac{1}{4}x\right)dx$

$= \frac{12}{7}\left[\frac{1}{4}x^4 + \frac{1}{8}x^2\right]_{x=0}^{x=1}$

$\therefore \underline{\mathbb{E}[X] = \frac{12}{7}\cdot\frac{3}{8} = \frac{9}{14} \simeq 0.64}$

$\mathbb{E}[Y] = \int_0^1 y\cdot f_Y(y)\,dy = \int_0^1 \frac{12}{7}\left(\frac{1}{3}y+y^4\right)dy$

$= \frac{12}{7}\left[\frac{1}{6}y^2 + \frac{1}{5}y^5\right]_{y=0}^{y=1}$

$\therefore \underline{\mathbb{E}[Y] = \frac{12}{7}\left(\frac{1}{6}+\frac{1}{5}\right) = \frac{12}{7}\cdot\frac{11}{30} = \frac{22}{35} \simeq 0.63}$

Now: $\mathbb{E}[X\cdot Y] = \int_0^1\int_0^1 xy\cdot\frac{12}{7}(x^2+y^3)\,dx\,dy$

$$= \frac{12}{7}\cdot\int_0^1 \left[\frac{1}{4}x^4 y + \frac{1}{2}x^2y^4\right]_{x=0}^{x=1} dy$$

$$= \frac{12}{7}\int_0^1 \frac{1}{4}y + \frac{1}{2}y^4\,dy$$

$$= \frac{12}{7}\left[\frac{1}{8}y^2 + \frac{1}{10}y^5\right]_{y=0}^{y=1}$$

$$\therefore \underline{\mathbb{E}[X\cdot Y] = \frac{12}{7}\cdot\frac{18}{80} = \frac{3\cdot 9}{7\cdot 10} = \frac{27}{70} \simeq 0.385}$$

Putting these together: $Cov(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\cdot\mathbb{E}[Y]$

$$= \frac{27}{70} - \frac{9}{14}\cdot\frac{22}{35}$$

$$\boxed{Cov(X,Y) = \frac{-9}{490} \simeq -0.018}$$

e) $\mathbb{E}[X^2] = \int_0^1 x^2 f_X(x)\,dx$

$$= \int_0^1 \frac{12}{7}\left(x^4 + \frac{1}{4}x^2\right)dx$$

$$= \frac{12}{7}\left[\frac{1}{5}x^5 + \frac{1}{12}x^3\right]_{x=0}^{x=1}$$

$$= \frac{12}{7}\left(\frac{1}{5} + \frac{1}{12}\right)$$

$$\underline{\mathbb{E}[X^2] = \frac{17}{35} \simeq 0.485}$$

$$\Rightarrow Var(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{17}{35} - \left(\frac{9}{14}\right)^2$$

$$\underline{Var(X) = \frac{71}{980} \simeq 0.072}$$

$$\mathbb{E}[Y^2] = \int_0^1 y^2\cdot f_Y(y)\,dy = \int_0^1 \frac{12}{7}\left(\frac{1}{3}y^2 + y^5\right)dy$$

$$= \frac{12}{7}\left[\frac{1}{9}y^3 + \frac{1}{6}y^5\right]_{y=0}^{y=1}$$

$$= \frac{12}{7}\left(\frac{1}{9} + \frac{1}{6}\right) = \underline{10/21}$$

$$\underline{\mathbb{E}[Y^2] = \frac{10}{21} \simeq 0.476}$$

$$\Rightarrow Var(Y) = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2$$

$$= \frac{10}{21} - \left(\frac{22}{35}\right)^2$$

$$\underline{Var(Y) = \frac{298}{3675} \approx 0.081}$$

So: $$\rho(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X) \cdot Var(Y)}}$$

$$= \frac{-9/490}{\sqrt{\frac{71}{980} \cdot \frac{298}{3675}}}$$

$$\boxed{\rho(X,Y) \approx -0.24}$$